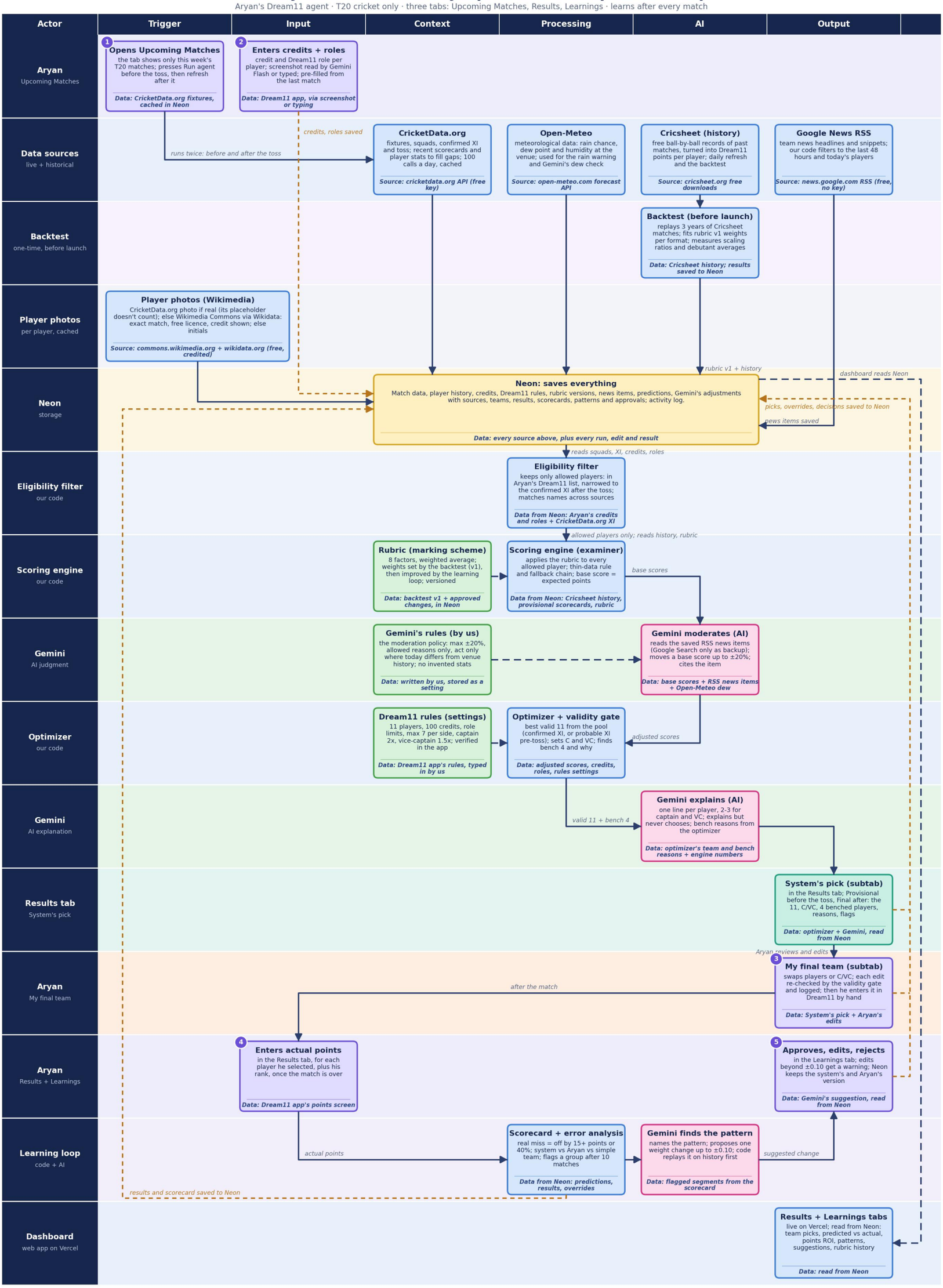
Fantasy Cricket Match Day Optimizer · Components Map

Aryan's Dream11 agent · T20 cricket only · three tabs: Upcoming Matches, Results, Learnings · learns after every match



What Aryan inputs himself

Five inputs; the numbers match the purple boxes above. Everything else is fetched, calculated and saved automatically.

	What he enters or does	Where and how	Time (estimate)	When
1	**Pick a match and press Run agent; press refresh after the toss**	Upcoming Matches tab	a few seconds	each match
2	**Credits and Dream11 roles**	Upcoming Matches tab: screenshot of the Dream11 player list read by Gemini Flash (he checks it), or typed; pre-filled from the last match	about 1 to 3 minutes	each match
3	**Edit the team if he disagrees (optional), then enter the final team in Dream11 by hand**	Results tab, My final team subtab; then the Dream11 app	his choice	each match
4	**Actual points for each player he selected, plus his rank**	Results tab, copied from Dream11's points screen	a few minutes	after each match
5	**Approve, edit or reject a suggested weight change**	Learnings tab	a minute	only when the system suggests one

Rarely: confirm a player name the system cannot match (once per player). Not Aryan's job: the rubric, Gemini's rules and the Dream11 rules settings are set up by us.

How the engine picks a team

Code filters and scores, Gemini moderates within limits, code builds a valid team, Aryan has the final say. All limits are guardrails we write; draft values until the rubric session.

Step 0 · Who is allowed	eligibility filter	Only players in Aryan's Dream11 list (credits and roles), narrowed to the confirmed XI after the toss. Names matched across sources; Aryan confirms an unmatched name once.
Step 1 · Base score	code, rubric weights	Weighted average of 8 factors: form (last 10, last 5 double), venue, opponent, current touch, opportunity, career, batting first or chasing, pitch type. Weights from the backtest; thin data blended with form; fallback chain for new players.
Step 2 · Gemini moderates	up to ±20% per player	Reads team news from Google News RSS (fetched and filtered by code); Google Search only as backup or for the pitch report. Only for injuries, a batting-order change, or pitch or dew that differs from venue history. Must cite the item; no news means no change.
Step 3 · Optimizer + gate	valid team only	Best valid 11 from the pool (confirmed XI after the toss; probable XI before it, from the last same-format match within 30 days). All limits from settings. Bench 4 with real reasons.
Step 4 · Captaincy + reasons	C highest, VC second	Captain = highest expected points, vice-captain = second highest, set by the optimizer. Gemini only explains: one line per player, 2-3 lines for captain and VC.
Step 5 · Aryan's final say	My final team subtab	Aryan can swap players or change C/VC; every edit is re-checked by the validity gate and logged with his reason. He enters the final team into Dream11 himself.
Scorecard	after every match	Compares system's pick, Aryan's final team and the simple team: prediction error per player, points ROI, efficiency vs the perfect team, captain hit, override hit rate.
Learning	10 matches, ±0.10	A real miss is off by 15+ points or 40%. A group is flagged after 10 matches of misses in one direction; Gemini proposes up to ±0.10; code replays it on history; Aryan approves, edits or rejects. Rain-shortened matches and mid-match injuries are skipped.

Every box names where its data comes from. Pink box = Gemini (AI). Purple box = Aryan. Green box = context written by us. Gold dashed line = saved into Neon. Navy dashed line = read only. Our backend code (not drawn) runs on Vercel and carries out each step.

Where every piece of data comes from

Data	Source	How it gets in	How often
Upcoming matches	**CricketData.org API**	our backend fetches the fixture list and caches it in Neon	once a day
Squads, playing XI, toss	**CricketData.org API**	fetched when Aryan presses Run agent, again after the toss	each run
Player history and past points	**Cricsheet.org downloads**	ball-by-ball loaded into Neon and converted to Dream11 points	daily refresh
Recent matches not yet on Cricsheet	**CricketData.org scorecards**	approximate points, marked provisional, replaced when Cricsheet publishes	as needed
New players with little history	**CricketData.org player stats**	rough estimate, flagged thin data	as needed
Weather (meteorological)	**Open-Meteo forecast API**	rain chance, dew point, humidity: rain warning and Gemini's dew check	each run
Team news	**Google News RSS**	fetched by our code, filtered to 48 hours and today's players, saved; Gemini reads the list	each run
Pitch report and backup news	**Google Search via Gemini 3.5 Flash**	only when RSS has nothing relevant; source links kept	when needed
Player photos	**Wikimedia Commons via Wikidata**	free licences only, exact person match, credit shown; else initials	once per player, 30-day refresh
Player credits and roles	**Dream11 app**	Aryan enters them (screenshot or typing); his list is the allowed list	each match
Dream11 rules and points table	**Dream11 app**	copied by us into settings at setup	when Dream11 changes
Actual points and rank	**Dream11 app**	Aryan types them into the Results tab	after each match
Rubric weights	**Backtest on Cricsheet**	v1 fitted per format; then changed only through approvals	setup + approvals
Gemini's rules	**written by us**	stored as settings, sent with every Gemini call	setup

What Neon saves, so the system can always say what happened

Match data	fixtures, venue, start time, squads, confirmed XI, toss result, weather and pitch notes
Player history	batting, bowling and fielding stats plus Dream11 points per player per match (Cricsheet and live)
Credits and roles	credit value and Dream11 role for each player in each match, entered by Aryan: the allowed list
Dream11 rules	credit budget, role limits, per-side cap, multipliers, points table: settings, verified in the app
Rubric versions	v1 from the backtest, then each approved change: weights, date, who approved: why the engine scored the way it did
Backtest outputs	past predictions vs actuals, format scaling ratios, debutant averages
Predictions	base and adjusted score per player, per run (draft or final), with the rubric version used
News items	every RSS headline Gemini was shown: title, snippet, link, time
Player photos	photo link, source (CricketData.org / Wikimedia / none), licence, author, credit link, skip reason
Gemini's adjustments	every move, its size, its reason and the news item it cites: proof of whether Gemini's judgment helps or hurts
Teams	System's pick: the 11, captain, vice-captain, the 4 benched players and the reasons shown to Aryan
Overrides	every swap or C/VC change in My final team, with Aryan's reason: who was right when they disagreed
Results	actual points per player and Aryan's rank, entered after the match
Scorecards	prediction error, points ROI, efficiency and captain hit per match
Patterns and suggestions	segment, average error, matches of evidence, Gemini's explanation, the system's suggestion and Aryan's version, approved, edited or rejected
Activity log	every run, entry and change with its time; rain-shortened matches flagged and excluded